EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Electronic Circuits (CSE251)
Mini Project

Project 1

A. Design and Analysis of a Diode-Based Rectifier Circuit.

Key Tasks

I. Rectifier Circuits
   a. Build a full-wave rectifier using a diode bridge (four diodes).
   b. Add smoothing capacitors and measure ripple voltage at the output.

II. Measurement and Analysis
   a. Use an AC signal generator and oscilloscope to observe input and output waveforms.
   b. Measure peak voltage, ripple voltage, and DC output voltage.
   c. Calculate rectification efficiency and analyze how diode forward voltage affects output.

III. Report
   a. Document circuit diagrams, measurements, waveform screenshots, and analysis.
   b. Discuss applications of diode rectification in real time.

Project 2

B. Design and Implementation of a BJT-Based AND Logic Gate

Key Tasks

I. Design the circuit
   a. Calculate base resistors for each transistor to ensure proper switching.
   b. Choose a collector load resistor for correct output voltage levels.

II. Simulate the circuit
   a. Verify correct AND truth table behavior with logic input combinations (00, 01, 10, 11).

III. Build the circuit
   a. Use common NPN transistors (e.g., 2N3904).
   b. Use a 5 V supply for logic HIGH levels.

IV. Test and measure
   a. Apply logic input combinations with switches or a signal generator.
   b. Measure output voltage levels and confirm logic operation.
   c. Analyze switching speed and power consumption.

IV. Report
   a. Document circuit diagrams, measurements, and analysis.
   b. Discuss applications of BJT-Based AND Logic Gate in real time.

Project 3

C. Design and Implementation of an Op-Amp Based Precision DC Voltage Amplifier.

Key Tasks

I. Circuit design
   a. Calculate resistor values for the desired gain (e.g., gain = 10 or 100).
   b. Design an offset null circuit using offset null pins if available, or by adding balancing resistors/potentiometers.

II. Simulation
   a. Verify gain and offset in SPICE simulation.

III. Construction
   a. Build on a breadboard with precision resistors (1% or better).

IV. Testing
   a. Use a low-noise DC voltage source and a high-precision multimeter to measure output voltage for known inputs.

V. Analysis
   a. Compare measured gain and offset to theoretical values.
   b. Plot output voltage vs temperature and discuss drift sources and compensation effectiveness.

VI. Report
   a. Document circuit diagrams, measurements, waveform screenshots, and analysis.
   b. Discuss applications of the Op-Amp Based Precision DC Voltage Amplifier in real time.

Grading rubric

- Design correctness and calculations: 40 pts
- Experimental procedure & data quality: 20 pts
- Discussion, error analysis & mitigation: 20 pts
- Documentation: 15 pts
- Report: 5 pts

Project covers the following:

Course Outcome: CO4
Program Outcome: PO1
Cognitive Level: C3
Psychomotor Level: P2, P3
Affective Level: A2
Knowledge Profile: K1, K3
Complex Engineering Problem: EP1, EP2